# Final Project Proposal: RGB-D Object Detection for Collision Avoidance

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Course: *Computer Vision (CS 5330)* – Professor: *Bruce Maxwell* – Due date: *July 21, 2026*

In our project, we will build a perception pipeline that converts an RGB-D video stream into a top-down occupancy map with real-world distances and scale for collision avoidance. For each frame, the system will detect and classify objects in the RGB image, place every obstacle in real-world 3D coordinates using aligned depth, remove the ground plane so the floor is not treated as an obstacle, and cluster the remaining 3D points so that obstacles the detector was never trained on are still detected. The output of this pipeline will be a Bird's Eye View (BEV) occupancy map labeling each obstacle's class and 3D position. We plan to evaluate the system on at least three datasets — the KITTI 3D object detection benchmark [1] for outdoor environments, SUN RGB-D [2] on indoor environments, and on our own recordings that we will collect using an RGB-D camera. SUN RGB-D and our collected dataset will come with real aligned depth measured by RGB-D sensors, while KITTI only offers RGB images with LiDAR measurements as the ground truth. Therefore, to obtain depth data for the KITTI dataset, we will use Depth Anything V2 [3] to predict depth, and for the datasets where we already have depth data, we will run variants using both Depth Anything V2 predictions and the depth data from the RGB-D sensor data. This one experiment will let us measure to what extent depth estimates from Depth Anything V2 can substitute real sensor data.

The pipeline will consist of six stages. First, a YOLO11 detector [4], pretrained on the COCO dataset [5], will produce 2D boxes and classes, with COCO's classes mapped to each dataset's labels. It will then back-project the depth map through the camera intrinsics into a 3D point cloud, serving as a LiDAR-like representation [6]. Next, the pipeline will use RANSAC [7] to remove the ground-plane as an object and a clustering algorithm to group remaining points into objects ourselves if needed. We will then fuse the YOLO object detections with the 3D clusters by projecting the clusters onto the image, so that each object gets a class label (or "unknown"), a 3D centroid, and a bounding box. Lastly, we will render obstacles on a BEV occupancy map grid with distance rings / shapes.

We will evaluate the success of our pipeline in various ways depending on the dataset. On KITTI, we will compute the the 3D centroid localization error against the dataset's 3D box annotations to measure how accurately we place objects and 3D average precision from the KITTI evaluation code to measure how reliably we find them. We'll compare both metrics to published pseudo-LiDAR results [6]. On SUN RGB-D, we will evaluate the same two metrics for the dataset's sensor depth – mean average precision under the community-standard 10-class protocol [8] and the centroid error. The COCO dataset only covers part of this 10-class list, so we will evaluate the classes that are scoreable, with the rest appearing as unknown obstacles. We may also interchange the nuScenes dataset [9] for the KITTI dataset as outdoor benchmarks depending on our preliminary results.

Beyond this core system, we have brainstormed possible extensions we could pursue depending on our initial progress and results by the checkin date. These extensions include recording our own data using an RGB-D camera with a UI that shows detected objects and collision alerts and recording a live demo, utilizing our pipeline to enable collision steering via object detection in the Habitat-Sim simulation environment [10] where a mobile robot navigates photorealistic indoor environments using the BEV map, finetuning the object detector on target domains, finetuning the metric depth head on KITTI's ground truth LiDAR depth measurements, and training a frustum-based 3D regression head [11], with the results reported as an ablation against the zero-shot baseline. Further, we will consider implementing our pipeline on a real robot and experimenting with 3D scene reconstruction methods, but this is likely out of scope. We are very excited about this project and look forward to hearing feedback and discussing preliminary results.

## References


[1] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in *CVPR*, 2012.

[2] S. Song, S. Lichtenberg, and J. Xiao, "SUN RGB-D: A RGB-D scene understanding benchmark suite," in *CVPR*, 2015.

[3] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth Anything V2," in *NeurIPS*, 2024.

[4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *CVPR*, 2016.

[5] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *ECCV*, 2014.

[6] Y. Wang, W.-L. Chao, D. Garg, B. Hariharan, M. Campbell, and K. Q. Weinberger, "Pseudo-LiDAR from visual depth estimation: Bridging the gap in 3D object detection for autonomous driving," in *CVPR*, 2019.

[7] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, vol. 24, no. 6, 1981.

[8] C. R. Qi, O. Litany, K. He, and L. J. Guibas, "Deep Hough voting for 3D object detection in point clouds," in *ICCV*, 2019.

[9] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuScenes: A multimodal dataset for autonomous driving," in *CVPR*, 2020.

[10] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Parikh, and D. Batra, "Habitat: A platform for embodied AI research," in *ICCV*, 2019.

[11] C. R. Qi, W. Liu, C. Wu, H. Su, and L. J. Guibas, "Frustum PointNets for 3D object detection from RGB-D data," in *CVPR*, 2018.